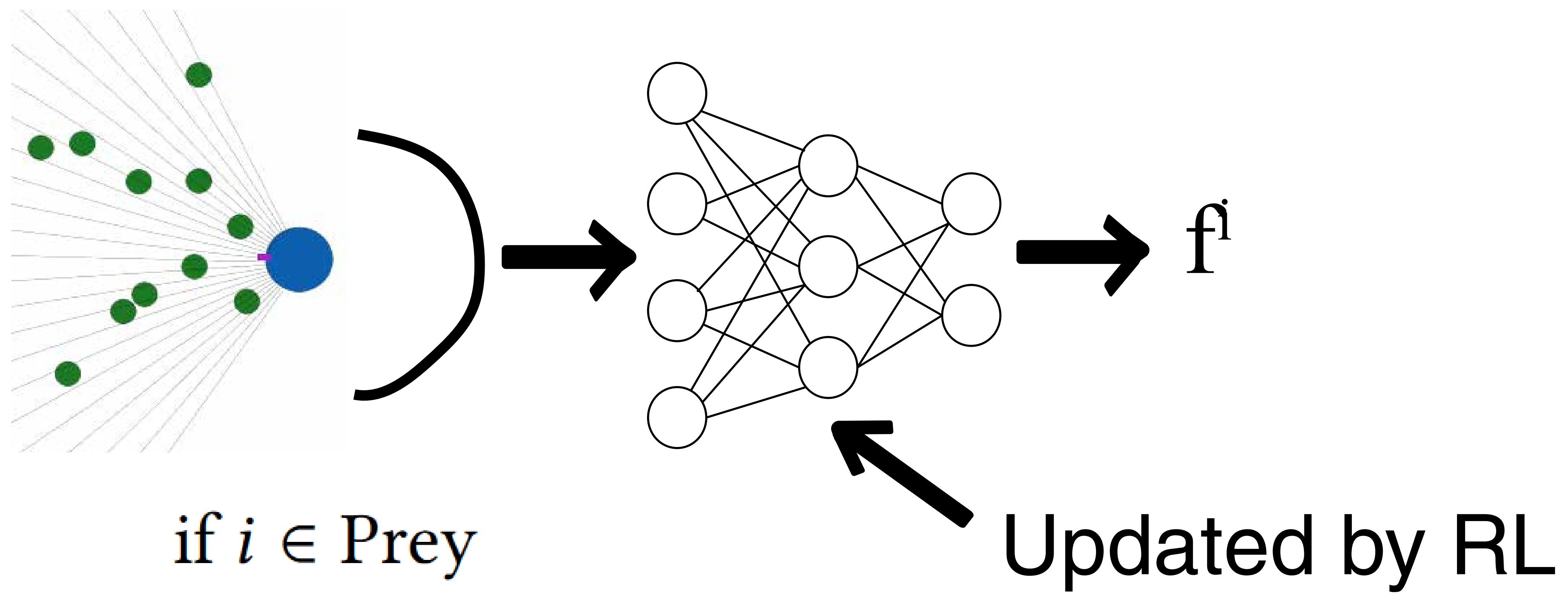
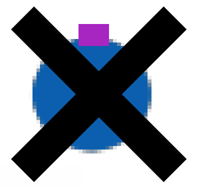
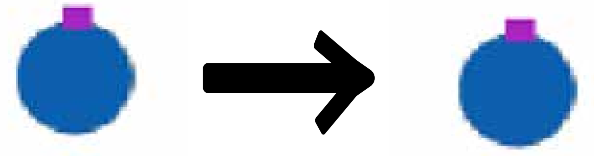
Simulation Procedure at Step j



1. Each agent i selects f^i from sensor info
2. Step physics simulation with motor acceleration
3. Update agents' energy by $\Delta e_j^i = \begin{cases} n_j^i - c_a\|f_j^i\| - c_b & \text{if } i \in \text{Prey} \\ \sum_{k \in \text{Prey eaten by } i} \eta e^k - d_a\|f_j^i\| - d_b & \text{if } i \in \text{Predator} \end{cases}$
4. Evaluate birth and death
   - An agent with age t and energy e die if e < 0 or with prob. h(t, e)



   - An agent with energy e reproduce with prob. b(e) with mutation



5. Compute r_j^i for each agent and (occasionally) update MLP by RL
6. Regenerate food items with maximum rate Δn

5. Compute r_j^i for each agent and (occasionally) update MLP by RL
6. Regenerate food items with maximum rate Δn

5. Compute r_j^i for each agent and (occasionally) update MLP by RL
6. Regenerate food items with maximum rate Δn

5. Compute r_j^i for each agent and (occasionally) update MLP by RL
6. Regenerate food items with maximum rate Δn